TECHNICAL TEST - QA

PT Logika Sarana Teknologi (Logique) — Bagian 1: Analytical & Test Design (35%)

Aplikasi Target:	Practice Software Testing (Toolshop)	**Target URL:**	https://practicesoftwaretesting.com
Lingkup Pengujian:	Register, Login, Edit Profile, Navigasi Menu	**Tanggal Pengujian:**	Oktober 2026
Kandidat QA:	QA Automation Engineering Candidate	**File Pendukung:**	01-test-design.xlsx, features/*.feature

1. Risk Analysis & Mitigation Matrix

Analisis risiko mengidentifikasi 7 area rentan dengan pembobotan **Likelihood (1–3) × Impact (1–3)**. Diurutkan dari skor tertinggi:

ID	Area Risiko	L (1-3)	I (1-3)	Skor	Level	Alasan Risiko & Strategi Mitigasi
R-01	**Autentikasi & Penyimpanan Sesi (Broken Auth / Session Hijacking)**	2	3	**6**	High	Jika token sesi tidak diamankan atau tidak di-invalidate saat logout/ganti password, akun rentan dibajak. **Mitigasi:** Verifikasi invalidasi token pasca logout dan proteksi private route.
R-02	**Validasi Batasan Usia Pendaftar (DOB Validation Bypass)**	3	2	**6**	High	Syarat usia 18–75 thn seringkali hanya dicek di browser sehingga rentan dibypass via direct API. **Mitigasi:** Terapkan Boundary Value Analysis (BVA) pada field tanggal lahir baik di UI maupun API.
R-03	**Persistensi Data Pembaruan Profil (Data Loss on Reload)**	2	3	**6**	High	Perubahan alamat/telepon gagal tersimpan permanen di database, memicu salah pengiriman pesanan. **Mitigasi:** Uji skenario update profil, reload browser (F5), dan verifikasi sinkronisasi payload backend.
R-04	**Ganti Password Tanpa Verifikasi Password Lama**	2	3	**6**	High	Jika sistem mengizinkan password baru tanpa verifikasi password lama yang cocok, akun mudah diambil alih. **Mitigasi:** Uji penolakan saat password lama salah dan uji kompleksitas password baru.
R-05	**Duplikasi Email saat Registrasi (Race Condition / Conflict)**	3	2	**6**	High	Registrasi dengan email yang sudah ada berisiko corrupt jika backend tidak memiliki unique constraint. **Mitigasi:** Uji registrasi dengan email terdaftar, validasi kode HTTP 422 dan pesan error UI.

ID	Area Risiko	L (1-3)	I (1-3)	Skor	Level	Alasan Risiko & Strategi Mitigasi
R-06	**Keamanan Input Form (Stored XSS / SQL Injection)**	2	3	**6**	High	Field nama atau alamat yang tidak disanitasi dapat mengeksekusi script jahat di profil atau invoice PDF. **Mitigasi:** Uji karakter HTML (<script>), petik, serta verifikasi output encoding di UI.
R-07	**Inkonsistensi Filter & Navigasi Kategori Produk**	2	2	**4**	Medium	Jika query slug navigasi putus atau state kategori corrupt, katalog gagal menampilkan daftar produk. **Mitigasi:** Uji routing navigasi, sinkronisasi URL slug, dan verifikasi relevansi produk.

2. Test Scenarios (Gherkin / BDD)

Skenario BDD disusun secara **declarative** (orientasi nilai bisnis, bukan teknis klik elemen). Tersimpan pada `features/auth_profile_navigation.feature`:

```
Feature: User Authentication, Profile Management, and Navigation
  As a customer of Toolshop
  I want to register, authenticate, manage my personal profile, and browse product categories
  So that I can securely manage my account and find tools easily

  Background:
    Given the customer is on the Toolshop application home page

  @auth @register @smoke @positive
  Scenario: Successful customer registration with valid personal details
    Given the customer navigates to the registration page
    When the customer submits the registration form with valid and complete personal information
    Then the system should register the new account successfully
    And the customer should be redirected to the login page with a success confirmation

  @auth @register @negative
  Scenario: Registration rejected when applicant is under legal age (under 18 years old)
    Given the customer navigates to the registration page
    When the customer attempts to register with a date of birth indicating age younger than 18 years
    Then the system should reject the registration submission
    And an age validation error stating that the customer must be at least 18 years old should be displayed

  @auth @login @data-driven
  Scenario Outline: Customer login attempts with various credential states
    Given the customer navigates to the login page
    When the customer attempts to sign in with email "<email>" and password "<password>"
    Then the authentication outcome should be "<status>"
    And the customer should receive the expected message "<expected_message>"

    Examples:
      | email                                | password         | status  | expected_message             |
      | customer@practicesoftwaretesting.com | welcome01        | SUCCESS | User dashboard displayed     |
      | customer@practicesoftwaretesting.com | WrongPassword99! | FAILED  | Invalid email or password    |
      | unregistered.user.2026@test.com      | ValidPass#2026   | FAILED  | Invalid email or password    |
      |                                      | ValidPass#2026   | FAILED  | E-mail is required           |
      | customer@practicesoftwaretesting.com |                  | FAILED  | Password is required         |

  @profile @positive
  Scenario: Successful update of customer profile details
    Given the customer is authenticated with an active account
    And the customer accesses the account profile settings
    When the customer updates the phone number and residential address with new details
    Then the updated profile should be saved successfully
    And the new information should persist upon reloading the profile page

  @profile @password @negative
  Scenario: Password change rejected when new password does not satisfy security complexity
    Given the customer is authenticated with an active account
    And the customer accesses the change password page
    When the customer attempts to set a weak new password lacking numbers or special characters
    Then the system should reject the password update request
    And a password complexity warning should prompt the customer to fulfill security requirements

  @navigation @catalog @positive
  Scenario: Targeted product catalog filtering through header category menu
    When the customer navigates to "Power Tools" via the top categories navigation menu
    Then the product catalog view should display only products classified under "Power Tools"
    And the page header and breadcrumb should display "Power Tools"
```

3. Detailed Test Cases Specification

Daftar 9 test case komprehensif (positif dan negatif) lengkap dengan precondition, test data, langkah uji, priority, dan teknik test design:

ID	Judul Test Case	Precondition & Test Data	Steps (Langkah Pengujian)	Expected Result	Priority	Teknik
TC-REG-01	**Registrasi user baru dengan data lengkap & valid**	**Pre:** Halaman home, guest. **Data:** John Doe, DOB: 1995-05-20, Email dinamis, Password@123	1. Klik Sign in -> Register. 2. Isi seluruh field dengan data valid. 3. Klik tombol Register.	Pendaftaran berhasil, otomatis redirect ke /auth/login, muncul pesan sukses.	High	Equivalence Partitioning
TC-REG-02	**Registrasi ditolak saat usia pendaftar < 18 tahun**	**Pre:** Form /auth/register. **Data:** DOB = 2008-10-05 (17 thn 364 hr), data lain valid	1. Isi form pendaftaran. 2. Masukkan DOB usia 17 thn. 3. Klik Register.	Pendaftaran ditolak, muncul error: pendaftar minimal harus berusia 18 tahun.	High	Boundary Value Analysis
TC-REG-03	**Registrasi ditolak saat email sudah terdaftar**	**Pre:** Email sudah aktif di DB. **Data:** customer@practicesoftwaretesting.com	1. Buka form registrasi. 2. Isi field email yang sudah terdaftar. 3. Submit form.	Sistem menolak registrasi, menampilkan feedback error: email sudah digunakan.	High	Error Guessing
TC-LOG-01	**Login berhasil dengan kredensial valid**	**Pre:** Akun aktif terdaftar. **Data:** customer@practicesoftwaretesting.com / welcome01	1. Buka /auth/login. 2. Masukkan email & password valid. 3. Klik tombol Login.	Otentikasi sukses, redirect ke /account, nama user tampil pada menu navigasi.	High	Decision Table
TC-LOG-02	**Login ditolak ketika password salah**	**Pre:** User terdaftar. **Data:** Email valid, Password: SalahPassword#99	1. Buka /auth/login. 2. Masukkan email valid & password salah. 3. Klik Login.	Login gagal, tetap di login page, muncul alert: 'Invalid email or password'.	High	Decision Table
TC-LOG-03	**Validasi field email & password dikosongkan**	**Pre:** Form login. **Data:** Email: "", Password: ""	1. Klik field email & password lalu biarkan kosong. 2. Klik Login.	Muncul pesan error: 'E-mail is required' & 'Password is required'. Form tidak terkirim.	Med	Equivalence Partitioning

ID	Judul Test Case	Precondition & Test Data	Steps (Langkah Pengujian)	Expected Result	Priority	Teknik
TC-PRF-01	**Update profil & verifikasi persistensi reload**	**Pre:** Login di /account/profile. **Data:** Phone baru, Alamat baru	1. Ubah nilai Phone & Address. 2. Klik Update Profile. 3. Reload browser page (F5).	Muncul notifikasi sukses update, data baru tetap tampil setelah halaman di-reload.	High	State Transition
TC-PWD-01	**Ganti password ditolak jika tidak memenuhi syarat**	**Pre:** Login di Change Password. **Data:** Current: welcome01, New: simplepass	1. Masukkan current pass valid. 2. Masukkan new pass tanpa angka/simbol. 3. Klik Save.	Sistem menolak update password, muncul peringatan syarat kompleksitas password.	High	Boundary & Decision
TC-NAV-01	**Navigasi menu kategori produk (Power Tools)**	**Pre:** Berada di beranda. **Data:** Kategori 'Power Tools' (/category/power-tools)	1. Klik dropdown Categories. 2. Pilih 'Power Tools'. 3. Verifikasi halaman katalog.	URL berubah ke /category/power-tools, judul 'Power Tools', hanya produk kategori tersebut tampil.	Med	State Transition

4. Penerapan Teknik Test Design Formal Secara Konkret

4.1. Teknik 1: Equivalence Partitioning (EP) & Boundary Value Analysis (BVA)

Diterapkan pada aturan bisnis field **Tanggal Lahir (Date of Birth)**: Batasan usia pendaftar wajib antara **18 s.d. 75 tahun** (Asumsi tanggal sistem pengujian: `04-10-2026`).

A. Tabel Partisi (Equivalence Partitioning):

Partisi	Rentang Nilai (Usia)	Contoh Nilai Uji (DOB)	Status	Expected Result
P1	Usia < 18 tahun (Minor)	2012-05-10 (14 thn)	Invalid	Ditolak: usia minimal pendaftar 18 tahun
P2	18 tahun ≤ Usia ≤ 75 tahun (Legal)	1990-08-15 (36 thn)	Valid	Diterima: registrasi dapat dilanjutkan
P3	Usia > 75 tahun (Senior di atas batas)	1945-01-01 (81 thn)	Invalid	Ditolak: usia maksimal pendaftar 75 tahun
P4	Tanggal di masa depan (Future Date)	2028-01-01 (-2 thn)	Invalid	Ditolak: tanggal lahir tidak boleh di masa depan
P5	Format non-tanggal / string acak	abc-def-ghij	Invalid	Ditolak: format tanggal wajib YYYY-MM-DD

B. Nilai Batas (Boundary Value Analysis):

Titik Uji	Nilai Uji (DOB)	Representasi Batas	Status	Expected Result
BVA-01	2008-10-05 (17 thn 364 hr)	Just below min boundary (18 thn - 1 hr)	Invalid	Ditolak (Error: usia < 18 thn)
BVA-02	2008-10-04 (Tepat 18 thn)	Exact minimum valid boundary	Valid	Diterima
BVA-03	2008-10-03 (18 thn 1 hr)	Just above min boundary (18 thn + 1 hr)	Valid	Diterima
BVA-04	1951-10-05 (74 thn 364 hr)	Just below max boundary (75 thn - 1 hr)	Valid	Diterima
BVA-05	1951-10-04 (Tepat 75 thn)	Exact maximum valid boundary	Valid	Diterima
BVA-06	1951-10-03 (75 thn 1 hr)	Just above max boundary (75 thn + 1 hr)	Invalid	Ditolak (Error: usia > 75 thn)

4.2. Teknik 2: Decision Table Testing (Logika Autentikasi Pengguna)

Kondisi Input & Aksi Sistem	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6
C1: Format Email Valid	Y	Y	Y	T	Y	T
C2: Email Terdaftar di Database	Y	Y	T	-	Y	-
C3: Password Sesuai Hash DB	Y	T	-	-	-	-
C4: Field Password Terisi	Y	Y	Y	Y	T	T
AKSI SISTEM (ACTIONS)						

Kondisi Input & Aksi Sistem	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6
A1: Login Sukses & Buat Token Sesi	X					
A2: Redirect ke Dashboard (/account)	X					
A3: Tampilkan Pesan 'Invalid email or password'		X	X			
A4: Tampilkan Validasi 'Format e-mail tidak valid'				X		
A5: Tampilkan Validasi 'Password is required'					X	X

4.3. Teknik 3: State Transition Testing (Siklus Akun Pengguna)

Current State	Trigger / Event	Action yang Dijalankan	Next State
S0: Guest	Submit Form Registrasi Valid	POST /users/register, simpan akun	**S1: Registered**
S0: Guest	Submit Kredensial Login Valid	POST /users/login, terbitkan token JWT	**S2: Authenticated**
S2: Authenticated	Ubah Nilai Profil & Klik Update	PUT /users/me, update database	**S3: Profile Updated**
S3: Profile Updated	Reload Browser (F5)	GET /users/me, populate form	**S3: Profile Updated (Persisted)**
S2: Authenticated	Klik Tombol 'Sign Out'	Hapus JWT session storage	**S0: Guest**

5. Out of Scope & Asumsi Pengujian

Out of Scope (Sengaja Tidak Dites)

- **Verifikasi Email SMTP Real Server:** Aplikasi demo langsung mengaktifkan akun tanpa tautan email nyata.
- **Payment Gateway Real Kartu Kredit:** Transaksi diuji dalam mode demo/dummy belanja e-commerce.
- **Social Login (OAuth Google / FB):** Arsitektur Toolshop belum menyediakan integrasi pihak ketiga.
- **Deep Penetration Testing (DAST):** Fokus pengujian adalah QA fungsional & test design sesuai brief.

Asumsi Pengujian

- **Environment Stabil:** Server demo publik `practicesoftwaretesting.com` beroperasi normal.
- **Isolasi Data Uji:** Registrasi otomatis menggunakan email unik berbasis timestamp agar bebas konflik duplikasi.
- **Basis Perhitungan Usia:** Pengecekan usia 18–75 thn berpedoman pada tanggal pengujian dijalankan.
- **Browser Standar:** Pengujian dioptimasi untuk browser modern berbasis Chromium (Chrome / Edge).